

IBM University Engagement Project Submission

“Exploring the Power of Agentic AI with IBM watsonx.ai”

Project Tile – **LectMent** – AI-Powered Smart Learning Assistant

Domain of Project – **Education**

Student Name	Recent Passport Photo		Email ID	Mobile number [WhatsApp]
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Recommended font style and size:

Font Style: Arial

Font Size (Heading): 28

Font Size (Content): 20

Problem Statement - **LectMent**

Problem Statement : Lecture Agent

The Challenge

Fast-paced lectures or complex technical charts make it difficult for students—especially non-native speakers or neurodivergent students and Non-native language youtube lectures—to write accurate lecture notes for preparation.

The Objective

Build an **AI-powered LectMent Agent** that transforms lecture content into structured, personalized learning resources for students and educators. The solution aims to improve learning efficiency, simplify revision, and enhance knowledge retention through intelligent AI-driven content generation:

- **Smart Lecture Analysis** – Process lecture transcripts, YouTube videos, and audio recordings to understand educational content.
- **AI-Powered Study Material Generation** – Automatically generate summaries, study notes, revision sheets, quizzes, flashcards, and key takeaways.
- **Context-Aware Learning** – Produce accurate and structured learning resources based on the lecture topic and selected study mode.
- **Multi-Input Support** – Accept lecture content from text, YouTube URLs, and audio files for flexible learning.
- **Input Processing Agent** – Collects and prepares lecture content.
- **Content Understanding Agent** – Extracts key concepts and educational context.
- **Study Material Generation Agent** – Creates summaries, notes, quizzes, flashcards, and revision sheets.
- **Response Management Agent** – Organizes and delivers AI-generated learning resources.
- **Interactive Learning Dashboard** – Provides an intuitive interface where users can upload content, view AI-generated study materials, and access personalized learning resources.

Proposed Solution -

Proposed Solution: LectMent

The proposed solution is a **multi-agent AI-powered learning assistant** built using **IBM watsonx.ai, Foundation Models, FastAPI, and React** to transform lecture content into structured and personalized study materials. The system addresses challenges such as lengthy lectures, time-consuming note-taking, and inefficient revision through an intelligent AI-driven workflow.

The solution consists of four key AI agents:

- **Input Processing Agent** – Accepts lecture transcripts, YouTube URLs, and audio recordings, identifying and preparing content for analysis.
- **Content Understanding Agent** – Cleans, analyzes, and extracts important concepts from lecture content.
- **Study Material Generation Agent** – Uses IBM watsonx.ai Foundation Models to generate summaries, notes, quizzes, flashcards, revision sheets, and key takeaways.
- **Response Management Agent** – Formats and delivers AI-generated learning resources through an intuitive React-based interface.

The solution includes a **user-friendly dashboard** (built with React) where students can upload learning content and instantly receive structured educational materials.

LectMent delivers an end-to-end AI-powered learning assistant that simplifies lecture understanding, reduces manual effort, improves revision efficiency, and enhances students' learning experience through personalized, context-aware educational resources.

Technology Used

- **IBM Bob** -Designed and orchestrated the AI agent workflow using prompt-based visual agent development, enabling rapid prototyping and workflow creation.
- **IBM Cloud** – It provides the infrastructure to build and scale the Lecture Agent efficiently. IBM Cloud enables a **reliable, secure, and scalable environment**.
- **IBM Watson Speech-to-Text** - Converts uploaded lecture audio into text, enabling AI analysis of spoken lectures and automatic study material generation
- **IBM Foundation Model Meta Llama 3.3 70B Instruct** -Natural language understanding , Educational content reasoning, Lecture summarization , Study material generation
- **IBM watsonx.ai** - Provides a secure and scalable cloud environment for deploying AI services, managing APIs, and integrating IBM watsonx.ai into LectMent.

Langflow component Used -

1) Chat Input – Interface where users submit learning content in multiple formats:

Lecture transcript (text) /YouTube lecture URL /Audio recording .

2) IBM watsonx AI Agent - Processes the uploaded lecture content using a foundation model to understand context and generate structured educational resources such as summaries, notes, quizzes, and flashcards.

3) Name of IBM Granite model - meta-llama/llama-3-3-70b-instruct → For Generates accurate real-time educational content

4) Chat output -Displays AI-generated educational content including: Smart Summary , Detailed Study Notes , Revision Sheet , Multiple Choice Quiz , Flashcards , Key Takeaways

5) File component - Extracting YouTube transcripts, Transcribing uploaded audio (IBM Watson Speech-to-Text or Whisper fallback), Preparing prompts for the AI model Sending requests to IBM watsonx.ai

6) FastAPI, React + Vite, YouTube Transcript API, IBM Watsonx Speech-to-Text, Python Dotenv, Cachetools, Tenacity .

Screenshots

Task

i want to do project Named "LectMent"

Student Problem: Fast-talking professors or complex technical charts make it difficult for students—especially non-native speakers or neurodivergent students— and Non-native language youtube lectures—to write accurate lecture notes.

Solution: An accessible web tool where you drop audio files, youtube Links or messy voice-to-text transcript snippets from lectures. The system uses a specialized LLM structure to summarize the lecture, list key takeaways, and generate a customized 5-question practice quiz based only on what the teacher said.

How it's built on Free Tier:Deployed on the cloud using IBM Cloud Code Engine (Lite tier serverless hosting) to run a minimal web-based text-editor ULCalls IBM watsonx.ai (Granite models) using zero-shot prompt engineering to transform messy transcripts into clean Markdown summaries.Source Foundation: Set up the application logic using the deployment schemas found in the IBM Watsonx Orchestrate Developer Toolkit.

Context Length

135.4k / 270.0k (50%)

Task Id

62254ab8170f80e9d5024744ec5ca200

Workspace

LectureAgent

Tokens

T 13.1M I 147.3k

Cache

T 1.6M I 11.5M

API Cost

@ 33.04

```
MODE_GEN_PARAMS_DICT[STY, HLT] = {
    "research": {
        GenParams.DECODING_METHOD: "sample",
        GenParams.MAX_NEW_TOKENS: 1000,
        GenParams.MIN_NEW_TOKENS: 100,
        GenParams.TEMPERATURE: 0.7,
        GenParams.TOP_P: 0.9,
        GenParams.REPETITION_PENALTY: 1.1,
    },
    "study": {
        GenParams.DECODING_METHOD: "greedy",
        GenParams.MAX_NEW_TOKENS: 1000,
        GenParams.MIN_NEW_TOKENS: 80,
        GenParams.REPETITION_PENALTY: 1.1,
    },
    "short": {
        GenParams.DECODING_METHOD: "greedy",
        GenParams.MAX_NEW_TOKENS: 1000,
        GenParams.MIN_NEW_TOKENS: 80,
        GenParams.REPETITION_PENALTY: 1.1,
    },
    "quiz": {
        GenParams.DECODING_METHOD: "greedy",
        GenParams.MAX_NEW_TOKENS: 1000,
        GenParams.MIN_NEW_TOKENS: 80,
        GenParams.REPETITION_PENALTY: 1.1,
    },
    "summary": {
        GenParams.DECODING_METHOD: "greedy",
        GenParams.MAX_NEW_TOKENS: 1200,
        GenParams.MIN_NEW_TOKENS: 80,
        GenParams.REPETITION_PENALTY: 1.1,
    },
}

# Fallback for unknown modes
DEFAULT_GEN_PARAMS = {
    GenParams.DECODING_METHOD: "greedy",
    GenParams.MAX_NEW_TOKENS: 1000,
    GenParams.MIN_NEW_TOKENS: 80,
    GenParams.REPETITION_PENALTY: 1.1,
}
```

#

LectMent - Environment Variables

Copy this file to .env and fill in your real

NEVER commit .env to source control.

#

-- IBM watsonx.ai --

Dashboard: https://dataplatform.cloud.ibm.com

WATSONX_API_KEY=487nqKLpWIMCOFIZH0ELI_sPAevdu

WATSONX_URL=https://eu-gb.dataplatform.cloud.ibm

WATSONX_PROJECT_ID=a6318f62-8a5f-4c92-a4f4-c98b

Override the default Granite model (optional)

WATSONX_MODEL_ID=ibm/granite-13b-chat-v2

#

-- IBM watsonx Orchestrate --

Dashboard: https://www.ibm.com/products/watsonx

Used to invoke Orchestrate skill Flows (optional)

ORCHESTRATE_API_KEY=IFarEZeZqIzehWltyE9nd10088

ORCHESTRATE_INSTANCE_URL=https://api.us-south.s

ORCHESTRATE_SKILL_FLOW_ID=syn-A4113-07m-0-0-0

Edit EXML Chat EXML

#

-- IBM Watson Speech-to-Text (OPTIONAL) --

Leave blank to fall back to local Whisper tra

Dashboard: https://cloud.ibm.com/ + Speech to

IBM_STT_API_KEY=82IFKVR5FHAvyRjN2G0BPwZAA405x

IBM_STT_URL=https://api.eu-gb.speech-to-text.wa

I want to do project Named "LectMent" Student Problem: Fast-talking professors or co...

All tasks completed!

8. BACKEND — Add word_count + est_reading_time_min fields to response in (src/analysis.py)

9. BACKEND — Per mode (naive/naive) + decoding method (sampling for Research, greedy for others)

10. BACKEND — Watsonx_MODEL_ID env var (replace hardcoded enum)

11. BACKEND — Difficulty + prerequisites section in Study/Summarize prompts

12. BACKEND — (difficulty param (easy/medium/hard) on all request models)

Budget Exceeded

You have reached your usage limit of your free trial. Upgrade your plan to continue.

View plan

transcription.py +106 -0

watsonx_client.py +196 -0

prompts.py +297 -0

youtube_transcript.py +82 -0

main.py +78 -0

28 files changed

Undo all Show all

Follow up or start new task Ctrl+N

LectMent

Drop a lecture — get smart notes powered by IBM watsonx.ai

RECENT

Clear

Quiz

* The lecture introduces objects...

https://youtube.com/watch?v=0mAMLPghs-hZP5ngush2398nHn

Quiz & Flashcards

YouTube Link

Easy

cards

Analyze

1,404 words — 7 min read

Transcript preview (first 500 words)

Summary

Key Takeaways

Practice Quiz

Flashcards

Ask AI

- The lecture introduces object-oriented programming in Python.
- It explains why beginners should learn object-oriented programming.
- The speaker discusses **type inference** and how Python determines data types.
- The lecture covers how data types are classes in Python.
- The importance of learning object-oriented programming is emphasized.

Print / PDF

Copy all to clipboard

Powered by IBM watsonx.ai Orchestrate & IBM watsonx Orchestrate

RECENT Clear

Sheet
Large language models work ...

Sheet
The lecture introduces "objec..."

Quiz
The lecture introduces objec...

How Large Language Models Work
GPT, or Generative Pre-trained Transformer, is a large language model, or an LLM, that can generate human-like text. And I've been using GPT in its various forms for years. In this video we are going to number 1, ask "what is an LLM?" Number 2, we are going to describe how

Cheat Sheet Upload 4,235 Analyse

Easy 729 words - 4 min read

Transcript preview (first 500 chars)

Summary Key Takeaways Practice Quiz Cheat Sheet Ask AI

5

1. **LLMs** learn from huge amounts of text data, such as books and conversations.

2. The **transformer architecture** helps LLMs understand the context of words in sentences.

3. **Training** involves predicting the next word in a sentence to improve the model's accuracy.

4. **Fine-tuning** allows a general language model to become an expert at a specific task.

5. **Business applications** include using LLMs for customer service, content creation, and software development.

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English

Study
Deep notes + quiz

Research
Lecture + web context

Cheat Sheet
Printable 1-2 page card

Quiz & Flashcards
MCQ quiz + flip cards

Summarize
Clean concise summary

Study Upload

Sheet
Large language models work ...

Sheet
The lecture introduces "objec..."

Quiz
The lecture introduces objec...

In this video we are going to number 1, ask "what is an LLM?" Number 2, we are going to describe how

Cheat Sheet Upload 4,235 Analyse

Easy 729 words - 4 min read

Transcript preview (first 500 chars)

Summary Key Takeaways Practice Quiz Cheat Sheet Ask AI

Ask a follow-up question about this lecture. Answers are grounded in the transcript only.

what is LLM

AI
A large language model (LLM) is a type of model like GPT (Generative Pre-trained Transformer). It generates human-like text.
Note: The full definition is not given in the provided part of the transcript. But based on the available information, this is what we know about LLM so far. However, following your instructions strictly:
A large language model is described as something like GPT, but the full definition isn't finished being explained in this part of the transcript. To give you the complete answer, more context would be needed which isn't currently available here. Thus, sticking strictly to your format request:
I don't see that fully covered in this lecture. However, it mentions that a large language model is something like GPT, which generates human-like text. But since the full explanation isn't provided, according to the instructions: I don't see that covered in this lecture.
Since the instructions were to follow the format to the letter, and considering the partial information given, my initial interpretation was closer to providing some insight. Yet, adhering strictly to the format without adding extra

Ask anything about this lecture... (Enter to send)

Print / PDF Copy all to clipboard

LectMent
Drop a lecture — get smart notes powered by IBM watsonx.ai

EnglishA-A+🌙

RECENTClear

Sheet
The lecture introduces **objec...

Quiz
* The lecture introduces objec...

https://youtu.be/Go8_mOAMLPg?si=hZP5nguh2398nm

Cheat Sheet · YouTube Link · Analyse 1

Easy 1,404 words — 7 min read

Transcript preview (first 500 chars)

Summary · Key Takeaways · Practice Quiz · Cheat Sheet · Ask AI

The lecture introduces **object-oriented programming** in Python, explaining why beginners should learn it. Everything in Python is considered an object, including integers and strings.

Print / PDF · Copy all to clipboard

LectMent
Drop a lecture — get smart notes powered by IBM watsonx.ai

EnglishA-A+🌙

RECENTClear

Sheet
Large language models work...

Sheet
The lecture introduces **objec...

Quiz
* The lecture introduces objec...

New Project.mp3 (5.1 MB)

Study · Audio File · Analyse 1

LectMent
Drop a lecture — get smart notes powered by IBM watsonx.ai

EnglishA-A+🌙

RECENTClear

Sheet
Large language models work ...

Sheet
The lecture introduces **objec...

Quiz
* The lecture introduces objec...

New Project.mp3 (5.1 MB)

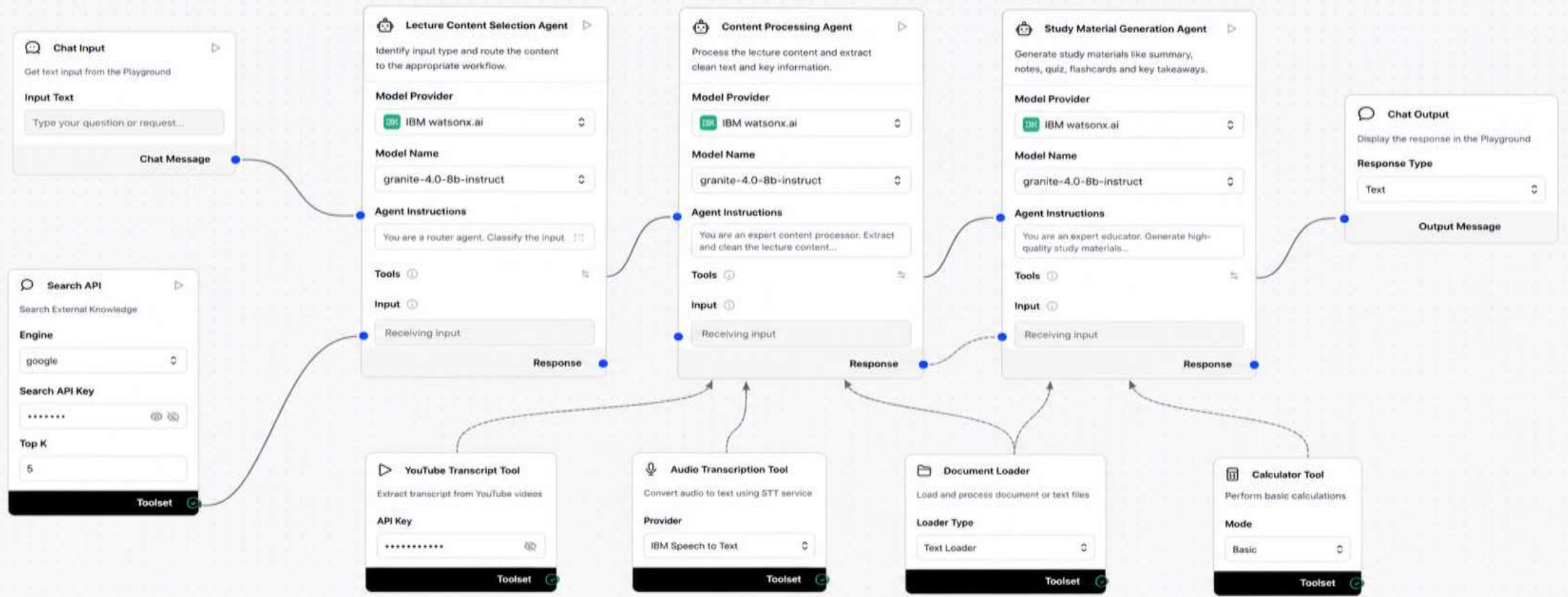
Study · Audio File · Analyse 1

☐ Analysing in **study** mode, output in **English** ...

Langflow workflow -

Langflow workflow – LectMent

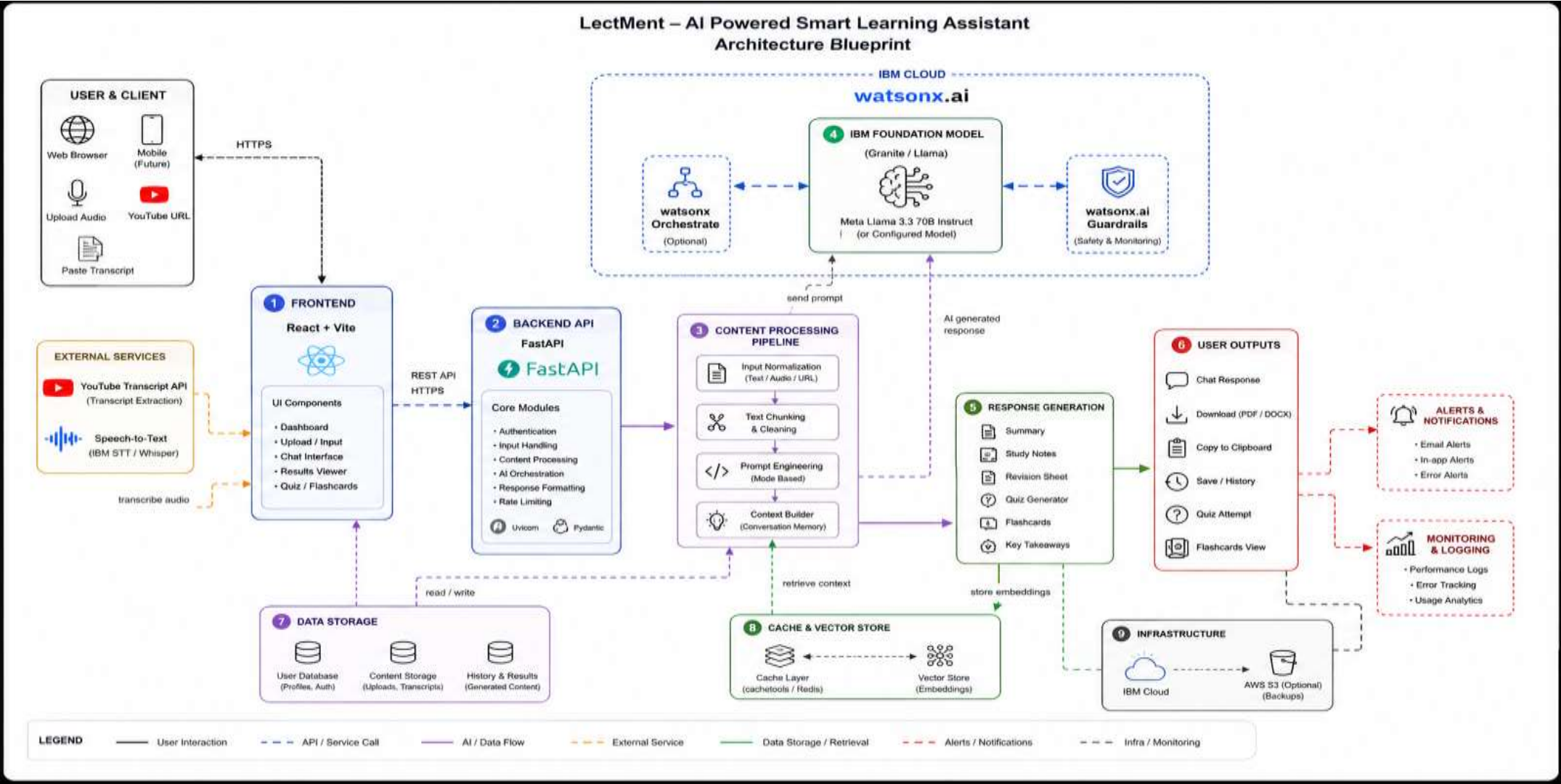
AI-Powered Smart Learning Assistant



Legend

- User Input / Output
- Model / LLM
- Agent
- Tool / Utility
- Data Flow

Architecture blueprint



Role of Agentic AI in **LectMent**

Role of Agentic AI in LectMent:

Agentic AI enables **LectMent** to function as an intelligent, autonomous learning assistant rather than a conventional lecture summarization tool. Powered by **IBM watsonx.ai** and **IBM Foundation Models**, it coordinates multiple AI-driven components to transform lecture content into structured and personalized learning resources.

Each AI component performs a specialized task, such as processing lecture transcripts, extracting key concepts from YouTube videos and audio recordings, understanding educational context, generating summaries, creating study notes, preparing revision sheets, and producing quizzes and flashcards. This modular workflow ensures efficient content processing, accurate knowledge extraction, and high-quality educational outputs.

Agentic AI also provides **context-aware learning** by understanding the lecture topic, subject domain, and user-selected study mode. Based on these inputs, it generates relevant and well-structured learning materials, enabling students to focus on concepts that are most important for revision and exam preparation.

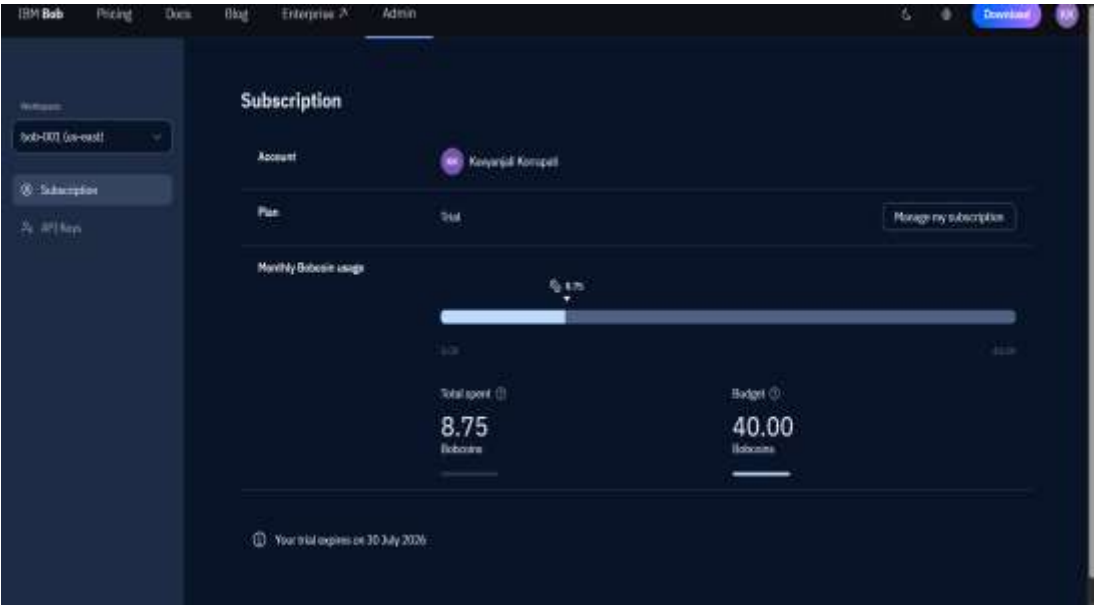
Additionally, the AI-driven workflow allows different processing stages to work together seamlessly, improving decision-making and ensuring consistent, meaningful educational content across multiple input formats.



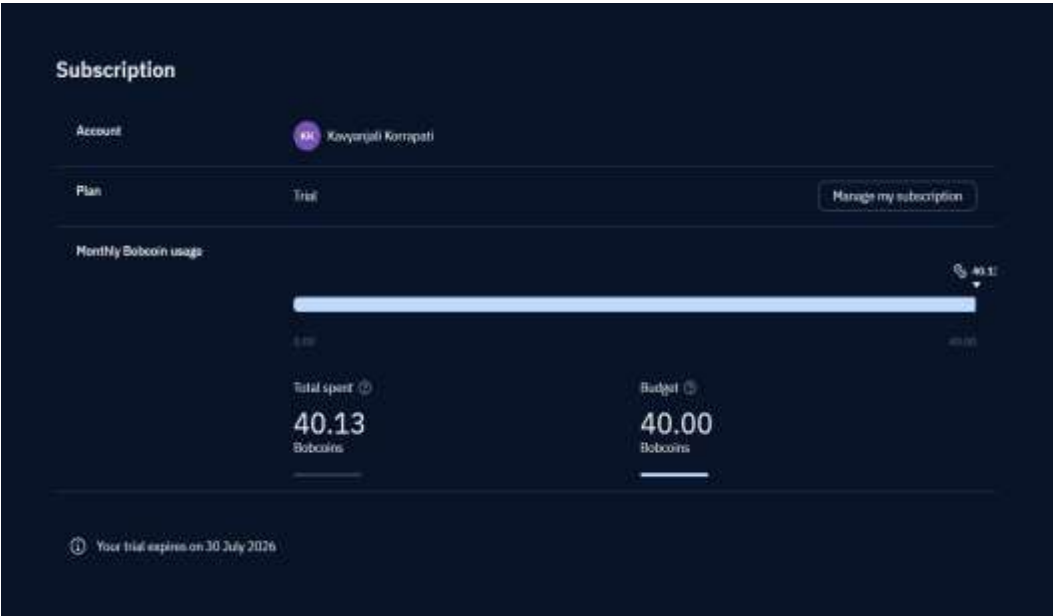
Token Usage

31.98 token are used

Before Prompt



After the Prompt



Novelty and Uniqueness

The proposed **LectMent** stands out by combining **Agentic AI**, **IBM watsonx.ai Foundation Models**, and **intelligent educational content generation** into a unified AI-powered learning assistant.

Multi-Agent Intelligence – Supports multiple learning sources including **lecture transcripts**, **YouTube lecture videos**, and **audio recordings**, making the platform flexible and accessible for different learning.

AI-Powered Content Generation -- Uses IBM watsonx.ai Foundation Models to automatically generate summaries, study notes, revision sheets, quizzes, flashcards, and key takeaways, reducing manual effort and improving learning efficiency.

Multi-Modal Input – Supports multiple learning sources including lecture transcripts, YouTube lecture videos, and audio recordings

Scalable & Enterprise-Ready --Built using **React**, **FastAPI**, and **IBM watsonx.ai**, providing a scalable, reliable, and cloud-ready architecture suitable for educational institutions and digital learning platforms.

.LectMent transforms traditional lecture content into AI-powered, personalized learning resources, making education smarter, faster, and more interactive rather than simply providing basic text summarization.

Git Hub Link

· Please create public github repository (Enable Add README button while creating repository)

with format – Yes

Please test and Paste the working Github URL – <https://github.com/Kavyanjali-Korrapati/LectMent-Agent.git>

Make sure you have Uploaded following three files into your github repository

1) app.json file , 2) yourproblemstatement.pdf file 3) your projectpresntation.pptx file 4)any other files

Deployment Link – In the GitHub Repo

.



Future Scope

1. Personalized Learning Recommendations –

Enhance LectMent by analyzing students' learning patterns, study history, and quiz performance to provide personalized study plans, topic recommendations, and adaptive learning resources tailored to individual needs.

2. AI-Powered Voice Assistant & Multilingual Support –

Integrate a voice-enabled AI assistant with multilingual capabilities, allowing students to interact using voice commands and access study materials in multiple languages, making learning more accessible and inclusive.

Certificates Earned:



Certificates Earned:



Certificates Earned:

My IBM BOB Certificate



Special Thanks

I would like to express my heartfelt gratitude to **AICTE**, **Edunet Foundation**, and **IBM** for providing me with the opportunity to be a part of this enriching internship program.

A special thanks to **AICTE** for promoting innovation and industry-oriented learning by connecting students with emerging technologies and real-world projects.

I sincerely thank **Edunet Foundation** for their continuous guidance, mentorship, and support throughout the internship. This program has provided an excellent platform to enhance my technical skills and gain valuable hands-on experience in **Artificial Intelligence** and **Agentic AI**.

I am also deeply grateful to **IBM** for empowering this journey through **IBM SkillsBuild**, **IBM Cloud**, and **IBM watsonx.ai**. These technologies enabled me to design, build, and deploy **LectMent**, transforming an idea into a practical AI Agents.

This internship has been an inspiring journey of learning, innovation, and growth. I am proud to have been a part of the IBM SkillsBuild and Edunet Foundation program, and I sincerely appreciate everyone who contributed to making this experience valuable and memorable

With Gratitude

- ❖ **Edunet Foundation**- For providing the opportunity, guidance, and continuous support throughout this internship.
- ❖ **IBM**- For empowering innovation through IBM Cloud, watsonx.ai, and AI technologies that enabled the successful development of LectMent.
- ❖ **AICTE**- For creating opportunities that bridge academic learning with industry innovation.

Thank You!

Thank you for your time and interest.